

Business Requirement Document (BRD)

Project Name: Real-Time Order Tracking & Communication Enhancement

System: Food Delivery Mobile Platform Document

Version:1.0

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1. Project Overview & Problem Statement

Executive Summary: This document outlines the business and functional requirements to introduce live GPS order tracking and automated push notifications within the food delivery application.

Problem Statement: Currently, customers lack real-time visibility into the order preparation and delivery stages. This ambiguity has led to a 35% increase in customer support tickets regarding estimated time of arrival (ETA) inquiries, reducing user satisfaction and straining operations.

Business Goals & Success Metrics:

- * Reduce "Where is my order?" customer support inquiries by 25% within 3 months of rollout.
- * Improve customer satisfaction (CSAT) rating for delivery experience from 3.8 to 4.5 out of 5.
- * Increase customer retention and repeat order frequency through enhanced transparency.

2. Project Scope

In-Scope:

- * Real-time GPS location tracking of the delivery rider on an interactive map.
- * Dynamic ETA calculation based on distance and live traffic conditions.
- * Automated push notifications and SMS alerts for major order status updates (Accepted, Preparing, Picked Up, Near Delivery, Completed).
- * In-app masked calling feature between customer and rider to protect contact privacy.

Out-of-Scope (Future Phases):

- * In-app live chat directly with the restaurant kitchen staff.
- * Multi-stop delivery batching visibility for customers.
- * Third-party courier tracking integrations.

3. Stakeholder Analysis

Stakeholders	Role	Responsibility & Impact
Customer	End User	Places orders, monitors status & receives food delivers.
Delivery Rider	Service Provider	Transmits real-time locations and complete order handoff.
Restaurant Partner	Merchant	Updates meal preparation status accurately on merchant portal.
Customer Support Team	Operations	Handles edge cases, delays & user escalations.
Product Team	Technical Implementation	Develops map APIs, backends sockets & UI interfaces

4. Functional Requirements (FR)

Order State Management: The system shall capture and update order states in real-time: Order Placed → Accepted → Preparing → Out for Delivery → Delivered

Live Geolocation Tracking: The mobile client shall render the rider's real-time GPS coordinates on an interactive map with an update interval of 10 seconds or less.

Dynamic ETA Engine: The system shall recalculate the Estimated Time of Arrival (ETA) dynamically using distance, speed, and real-time traffic data via map APIs.

Automated Notifications: The system shall trigger push notifications and SMS within 5 seconds of any state transition to keep the user informed asynchronously.

Masked Communication: The system shall provide an in-app VoIP/telephony bridge that allows the customer and delivery rider to contact each other without exposing personal phone numbers.

5. Non-Functional Requirements (NFR)

Performance & Latency: Map tile rendering and live GPS coordinate refreshes must complete within under 2 seconds over standard 4G mobile networks.

Scalability & Concurrency: The tracking infrastructure must support up to 10,000 concurrent active tracking sessions during peak meal hours (e.g., 12:00 PM – 2:00 PM and 7:00 PM – 9:30 PM) without latency degradation.

Security & Data Privacy: All GPS coordinates and personal identifiable information (PII) must be encrypted in transit using TLS 1.3 and at rest using AES-256 standards.

High Availability: The order tracking service must maintain 99.9% uptime to prevent operational blindness during peak trading periods.

6. Agile User Stories & Acceptance Criteria

User Story 01: Interactive Map Tracking

As a customer,

I want to observe my delivery partner moving on a live map with an accurate ETA,

So that I know exactly when to receive my order without anxiety.

Acceptance Criteria (Gherkin Format):

Scenario: Successful real-time tracking during transit

Given an order status is marked as Out for Delivery,

When the customer opens the active tracking view,

Then the map displays the rider pin, delivery address pin, and polyline route,

And the ETA displays remaining minutes refreshed every 60 seconds.

User Story 02: Private Contact Channel

As a Delivery Rider,

I want to contact the customer directly via an in-app calling mechanism,

So that I can clarify location details without sharing or seeing private phone numbers.

Acceptance Criteria (Gherkin Format):

Scenario: Masked call initiation

Given the order is in progress,

When either party taps the Call button,

Then the telephony server bridges the call using virtual proxy numbers,

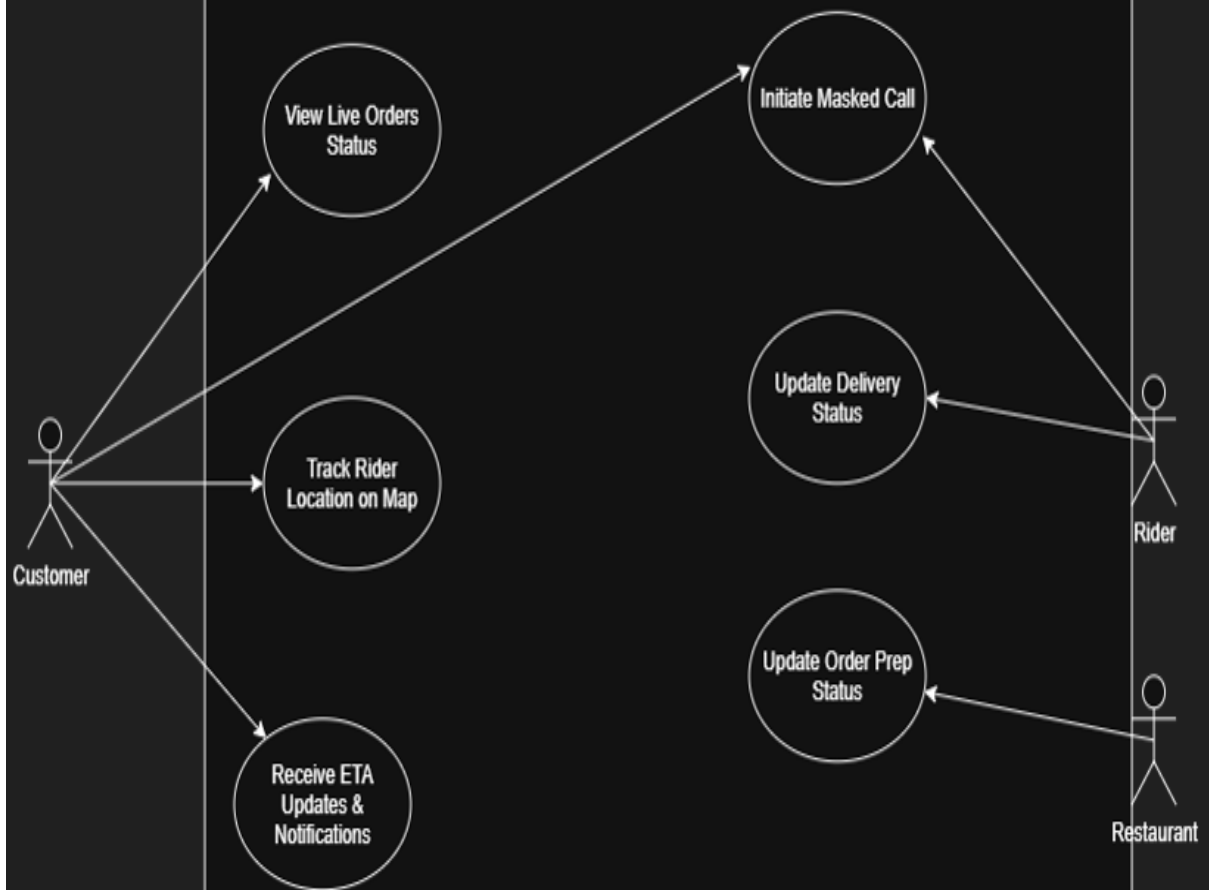
And no raw personal telephone numbers are exposed on the device screen.

7. Visual System Models & Diagrams

7.1) System Use Case Diagram

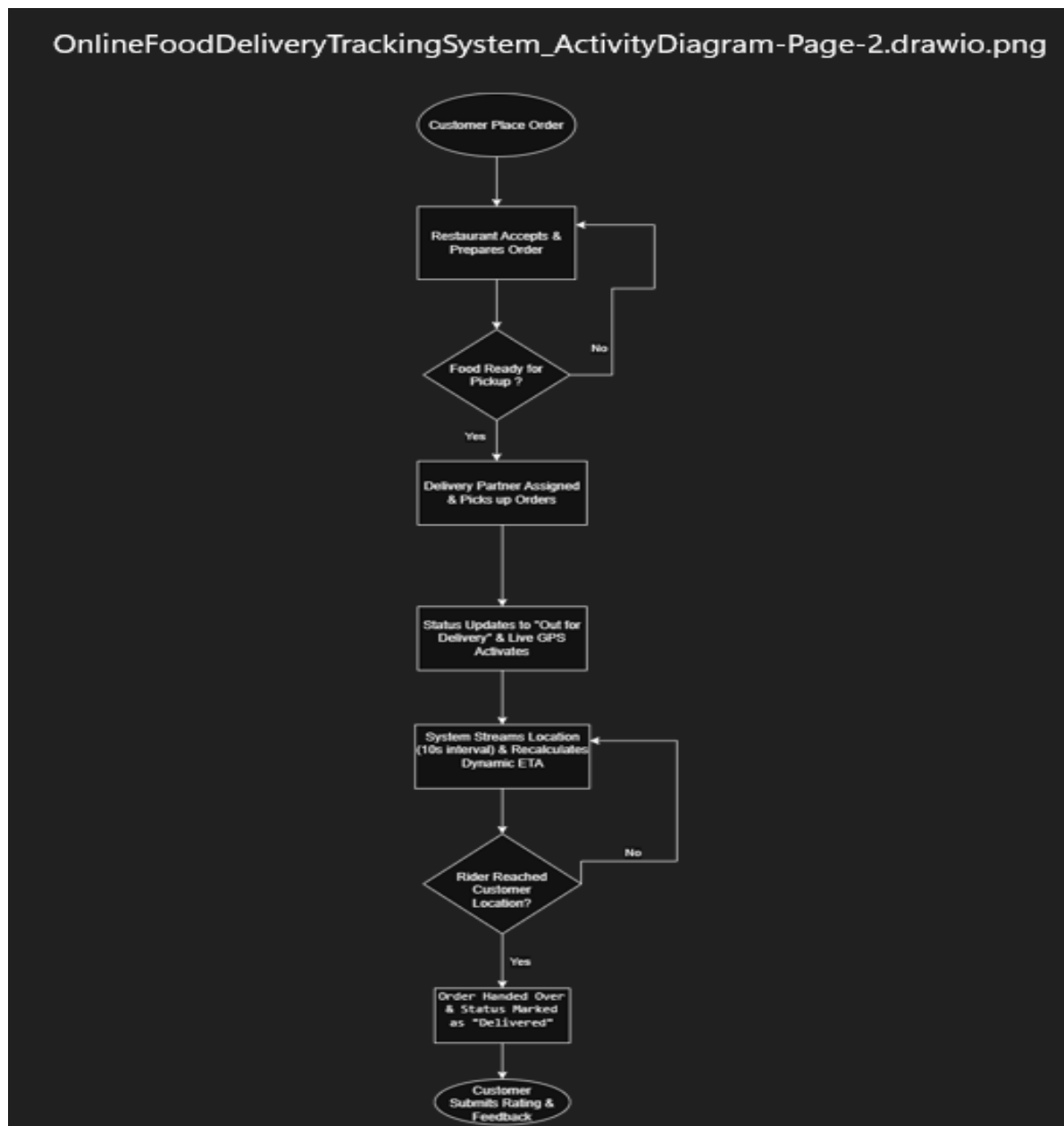
Description: Illustrates the functional boundary of the tracking system and the interactions between primary actors (Customer, Delivery Rider, Restaurant Partner) and system capabilities.

Food Delivery Order Tracking System



7.2) End-to-End Order Tracking Process

Description: Details the step-by-step decision points and automated workflow from initial order acceptance to meal delivery and customer feedback collection.



8. Assumptions, Constraints & Dependencies

Assumptions:

Customers and Delivery Riders have smartphones with active internet connectivity and GPS services enabled.

Restaurant partners maintain real-time accuracy when updating kitchen prep statuses on the merchant portal.

Constraints:

GPS tracking accuracy is subject to cellular network strength and satellite reception in urban and remote zones.

Third-party telephony API availability governs the uptime of the masked calling functionality.

Dependencies:

Integration with third-party geolocation mapping APIs (e.g., Google Maps Platform / Mapbox) for polyline routing and ETA computation.

Cloud messaging gateways (Firebase Cloud Messaging / Twilio SMS) for real-time push alerts.

9. Risk Analysis & Mitigation

Risk Description	Severity	Likelihood	Mitigation Strategy
GPS Signal Drift / Battery Drain: Continuous location streaming drains rider device battery rapidly.	High	High	Implement adaptive polling (e.g., reduce update frequency to 30s when moving at uniform speeds).
Data Privacy Breach: Leakage of customer delivery addresses or contact numbers during live sessions	Critical	Low	Enforce strict end-to-end encryption (TLS 1.3/AES-256) and ephemeral masked virtual routing for phone numbers.
High Traffic Spike Latency: Server delay during rush-hour meal times (12:00 PM – 2:00 PM).	High	Medium	Utilize autoscaling microservices and WebSocket connections for lightweight location streaming.